

ME

1. What is the primary purpose of scientific research?

- A) To prove personal opinions
- B) To generate reliable, valid knowledge
- C) To support existing beliefs
- D) To entertain the public

2. Which is a characteristic of a good research problem?

- A) Ambiguity
- B) Simplicity
- C) Specificity
- D) Subjectivity

3. What is the first step in the scientific research process?

- A) Conducting experiments
- B) Reviewing the literature
- C) Formulating a hypothesis
- D) Analyzing data

4. Which best describes a hypothesis?

- A) A proven fact
- B) A testable statement
- C) A random guess
- D) An experimental result

5. What is the primary purpose of a literature review?

- A) To gather raw data
- B) To identify gaps in existing research
- C) To formulate final hypotheses
- D) To draw conclusions

6. Which method gathers observable, empirical, and measurable evidence systematically?

- A) Qualitative research
- B) Quantitative research
- C) Theoretical research
- D) Philosophical research

7. What is an independent variable?

- A) A variable that is measured
- B) A variable that is manipulated
- C) A variable kept constant
- D) A variable dependent on others

8. Which is NOT a characteristic of qualitative research?

- A) Subjective analysis
- B) Use of open-ended questions
- C) Statistical analysis
- D) Exploratory nature

9. Which sampling method divides the population into subgroups and samples each?

- A) Simple random sampling
- B) Systematic sampling
- C) Stratified sampling
- D) Convenience sampling

10. The consistency of a study or test is called:

- A) Validity
- B) Reliability
- C) Accuracy
- D) Precision

11. What is a control group?

- A) The group that receives the treatment
- B) The group used to measure the treatment effect
- C) A group not in the study
- D) The group that designs the experiment

12. Which is a qualitative data collection method?

- A) Surveys
- B) Experiments
- C) Interviews
- D) Quasi-experiments

13. In research, 'bias' refers to:

- A) The precision of measurements
- B) A systematic error that skews results
- C) A random error
- D) The validity of an experiment

14. What is the purpose of a double-blind study?

- A) To increase sample size
- B) To prevent bias
- C) To decrease cost
- D) To simplify procedures

15. What is 'operationalization'?

- A) Defining variables in practical, measurable terms
- B) Conducting the research
- C) Analyzing data
- D) Reviewing literature

16. 'Statistical significance' implies that:

- A) Results are due to chance
- B) Results are due to chance
- C) Results are important
- D) Results are unlikely due to chance

17. Which design best determines causality?

- A) Descriptive
- B) Correlational
- C) Experimental
- D) Exploratory

18. NOT a quantitative data collection method:

- A) Questionnaires
- B) Structured interviews
- C) Focus groups
- D) Surveys

19. Main goal of a pilot study:

- A) Complete the full study
- B) Test feasibility of design
- C) Analyze data
- D) Review literature

20. Research that seeks to understand phenomena in natural settings

- A) Quantitative
- B) Qualitative
- C) Experimental
- D) Theoretical

21. A case study is:

- A) A study with a large sample
- B) An in-depth analysis of a single/few cases
- C) A study using only statistics
- D) Purely theoretical

22. The extent to which results generalize to other people/settings:

- A) Internal validity
- B) External validity
- C) Construct validity
- D) Content validity

23. Exploratory research mainly:

- A) Tests specific hypotheses
- B) Establishes causality
- C) Explores under-researched areas
- D) Uses large samples

24. In experiments, which variable is influenced by the independent variable?

- A) Control variable
- B) Dependent variable**
- C) Extraneous variable
- D) Confounding variable

25. Triangulation in qualitative research is:

- A) Using multiple data sources/methods to verify findings**
- B) Using statistics to validate results
- C) Using control groups
- D) Testing hypotheses experimentally

26. A common qualitative data analysis method:

- A) Regression analysis
- B) Content analysis**
- C) ANOVA
- D) Chi-square test

27. Intensive study of an individual/group/event:

- A) Survey research
- B) Experimental research
- C) Case study research**
- D) Cross-sectional research

28. NOT an ethical consideration:

- A) Informed consent
- B) Confidentiality
- C) Subjectivity**
- D) Avoiding harm

29. Main advantage of longitudinal studies:

- A) Less expensive
- B) Smaller samples
- C) Study changes over time**
- D) Quick results

30. 'Peer review' means:

- A) Review by friends/colleagues
- B) Evaluation by field experts**
- C) Review by participants
- D) Evaluation by funders

31. Research that manipulates variables to establish cause-effect:

- A) Descriptive
- B) Experimental**
- C) Correlational
- D) Qualitative

32. Primary purpose of the null hypothesis:

- A) Predict a specific outcome
- B) Be proven true
- C) Be tested and possibly rejected
- D) Support the research hypothesis

33. Mixed-methods research:

- A) Only quantitative
- B) Only qualitative
- C) Combines qualitative and quantitative
- D) Uses only secondary data

34. Key characteristic of action research:

- A) Controlled experiment
- B) Solves an immediate, practical problem
- C) Focus on theory only
- D) Large samples

35. Ability of a test to measure what it should:

- A) Reliability
- B) Validity
- C) Objectivity
- D) Subjectivity

36. NOT a type of observational research:

- A) Naturalistic observation
- B) Participant observation
- C) Laboratory observation
- D) Experimental observation

37. Primary focus of descriptive research:

- A) Establish causality
- B) Explore new phenomena/ new population
- C) Describe characteristics of a population/phenomenon
- D) Test hypotheses

38. Best test for comparing means of two independent groups:

- A) Chi-square
- B) t-test
- C) ANOVA
- D) Correlation

39. Interview with fixed, non-deviating questions:

- A) Structured interview
- B) Semi-structured interview
- C) Unstructured interview
- D) Informal interview

40. Key feature of ethnographic research:

- A) Short-term data collection
- B) Use of experiments
- C) In-depth study of cultural groups
- D) Statistical analysis

41. Which of the following is an example of a secondary data source?

- A) Survey responses
- B) Interview transcripts
- C) Government reports
- D) Experimental results

42. What does 'random assignment' mean in experimental research?

- A) Randomly selecting participants from the population
- B) Randomly assigning participants to different groups
- C) Randomly choosing variables to study
- D) Randomly collecting data

43. Which type of validity concerns generalization to real-world settings?

- A) Internal validity
- B) Construct validity
- C) Ecological validity
- D) Criterion validity

44. What is a meta-analysis?

- A) Analyzing a single study in depth
- B) Combining results from multiple studies to draw a general conclusion
- C) Reviewing literature on a topic
- D) Conducting original research

45. Which is a primary data source?

- A) Census data
- B) Literature reviews
- C) Direct observations
- D) Historical records

46. Which type of research aims to generate new theories?

- A) Applied research
- B) Basic research
- C) Experimental research
- D) Descriptive research

47. What is the role of an Institutional Review Board (IRB)?

- A) To fund research projects
- B) To review and approve research involving human subjects
- C) To conduct the research
- D) To publish findings

48. A common purpose of a focus group is to:

- A) Conduct a large-scale survey
- B) Gather in-depth qualitative data**
- C) Perform statistical analysis
- D) Manipulate variables in an experiment

49. 'Informed consent' means:

- A) Participants' confidentiality keeping procedures and rules
- B) Participants agree to take part after being fully informed**
- C) Researchers' commitment to ethical guidelines
- D) Review by an ethics committee

50. The purpose of control variables is to:

- A) Test the hypothesis
- B) Ensure reliability
- C) Eliminate influence of extraneous factors**
- D) Measure the dependent variable

51. Which method studies the same subjects over a long period?

- A) Cross-sectional research
- B) Longitudinal research**
- C) Experimental research
- D) Case study research

52. A confounding variable is:

- A) Kept constant both independent and dependent variables
- B) Influences both independent and dependent variables, leading to spurious effects**
- C) Manipulated
- D) Measured as the outcome

53. What is the primary file format Stata uses for datasets?

- A) .csv
- B) .xlsx
- C) .dta**
- D) .sav

54. Which R function reads a CSV file?

- A) read_csv
- B) read.csv**
- C) csv.read
- D) readfile.csv

55. In Stata, which command creates a new variable?

- A) gen**
- B) newvar
- C) create
- D) addvar

56. In RStudio, which package is commonly used for data manipulation?

- A) ggplot2
- B) dplyr**
- C) caret
- D) shiny

57. How do you install a package in R?

- A) `package.install("name")`
- B) `install.packages("name")`**
- C) `library("name")`
- D) `install("name")`

58. Which Stata command lists all variables in the dataset?

- A) `variables`
- B) `list`
- C) `describe`**
- D) `vars`

59. What symbol usually denotes the null hypothesis?

- A) H_1
- B) H_2
- C) H_0**
- D) H_{null}

60. Which R function performs a t-test?

- A) `t.test()`**
- B) `test.t()`
- C) `ttest()`
- D) `test.t`

61. In Stata, which command performs a paired t-test?

- A) `ttest var1, by(var2)`
- B) `ttest var1==var2`**
- C) `ttestpaired var1 var2`
- D) `pairtest var1 var2`

62. The significance level is typically denoted as:

- A) α**
- B) β
- C) p
- D) μ

63. If $p\text{-value} < \alpha$, you should:

- A) Accept the null hypothesis
- B) Reject the null hypothesis**
- C) Accept the alternative hypothesis
- D) Fail to reject the null hypothesis

64. Which R function performs a chi-squared test?

- A) `chi.test()`
- B) `chisq.test()`
- C) `chisquare()`
- D) `chi_square()`

65. Which R package is commonly used to create dynamic reports?

- A) shiny
- B) `rmarkdown`
- C) tidyverse
- D) knitr

66. In Stata, which command can export tables to LaTeX?

- A) `texsave`
- B) `estout`
- C) `outreg2`
- D) `esttab`

67. Which function creates a PDF report from an R Markdown file?

- A) `rmarkdown::render()`
- B) `pdf::render()`
- C) `markdown::pdf()`
- D) `render::rmarkdown()`

68. Which format is NOT supported as an R Markdown output document?

- A) HTML
- B) Word
- C) PDF
- D) PNG

69. What is the primary purpose of the knitr package?

- A) Creating visualizations
- B) Data manipulation
- C) Dynamic report generation
- D) Statistical modeling

70. In Stata, which command creates a table of summary statistics?

- A) `summarize`
- B) `tabstat`
- C) `descriptive`
- D) `sumtable`

71. Which R package is most commonly used for graphics?

- A) `ggplot2`
- B) `dplyr`
- C) `data.table`
- D) `caret`

72. Which Stata command generates a scatter plot?

- A) plot
- B) scatter
- C) graph scatter
- D) plot scatter

73. Which R function creates a bar plot?

- A) barplot()
- B) plot.bar()
- C) bar.plot()
- D) plotbar()

74. In ggplot2, aes() specifies:

- A) Aesthetic mappings
- B) Data filters
- C) Color schemes
- D) Graph titles

75. How do you create a histogram in Stata?

- A) histogram varname
- B) graph hist varname
- C) plot histogram varname
- D) hist varname

76. Which R function creates a boxplot?

- A) boxplot()
- B) plot.box()
- C) box.plot()
- D) plotbox()

77. What is a Type I error?

- A) Rejecting a true null hypothesis
- B) Accepting a false null hypothesis
- C) Accepting a true null hypothesis
- D) Rejecting a false null hypothesis

78. What is a Type II error?

- A) Rejecting a true null hypothesis
- B) Accepting a false null hypothesis
- C) Accepting a true null hypothesis
- D) Rejecting a false null hypothesis

79. Having too small a sample size primarily leads to:

- A) Increased Type I error rate
- B) Decreased power
- C) Increased effect size
- D) Reduced measurement error

80. Multicollinearity occurs when:

- A) Independent variables are too highly correlated
- B) The dependent variable is not normal
- C) Residuals are heteroscedastic
- D) The model is overfitted

81. In Stata, how can you test for heteroscedasticity?

- A) `hettest`
- B) `test hetero`
- C) `homoscedasticity`
- D) `residualtest`

82. Overfitting in regression means the model:

- A) Is too complex and fits noise
- B) Is too simple
- C) Is incorrectly specified
- D) Has high variance and low bias

83. In R, which function best checks multicollinearity?

- A) `cor()`
- B) `vif()`
- C) `multi()`
- D) `collinear()`

84. Which Stata command identifies influential data points?

- A) `influence`
- B) `outlier`
- C) `dfbeta`
- D) `leverage`

85. Which R package provides tools for advanced regression modeling?

- A) `lmtools`
- B) `caret`
- C) `regression`
- D) `advancedreg`

86. Which Stata command runs logistic regression?

- A) `logit`
- B) `reglog`
- C) `logreg`
- D) `logisticreg`

87. Which R function fits a linear model?

- A) `linear.model()`
- B) `lm()`
- C) `fit.model()`
- D) `linmod()`

88. Which Stata command merges datasets by keys?

- A) append
- B) combine
- C) merge
- D) join

89. In R, how do you merge two data frames by a common column?

- A) merge(df1, df2)
- B) merge(df1, df2, by = "column")
- C) join(df1, df2)
- D) merge.dataframes(df1, df2)

90. Which Stata command checks for missing values?

- A) misschk
- B) misstable
- C) checkmiss
- D) missing

91. In R, which function provides a basic summary that can reveal NAs?

- A) summary.missing()
- B) na.summary()
- C) summary()
- D) missing()

92. Which Stata command generates descriptive statistics?

- A) summarize
- B) describe
- C) stats
- D) summary

93. What is the null hypothesis (H0) in hypothesis testing?

- A) The hypothesis the researcher is trying to prove
- B) There is no effect or no difference
- C) The alternative hypothesis
- D) A hypothesis that cannot be tested

94. What is the alternative hypothesis (H1)?

- A) The hypothesis assumed true initially
- B) There is no effect or difference
- C) There is an effect or difference
- D) Always results in a Type I error

95. A Type I error occurs when:

- A) Accepting H0 when false
- B) Rejecting H0 when true
- C) Accepting H1 when false
- D) Rejecting H1 when true

96. The power of a test is the probability of:

- A) Rejecting H_0 when it is true
- B) Rejecting H_0 when it is false
- C) Accepting H_0 when it is false
- D) Making a Type I error

97. Which is a parametric test?

- A) Chi-square test
- B) Mann-Whitney U test
- C) t-test
- D) Kruskal-Wallis test

98. A z-test is typically used when:

- A) Sample sizes are small
- B) The population standard deviation is known
- C) Data are ordinal
- D) Comparing more than two groups

99. What is a critical value?

- A) The test statistic value separating rejection from non-rejection regions
- B) Max allowable Type II error
- C) $P(\text{data} | H_0 \text{ true})$
- D) The mean under H_0

100. A two-tailed test evaluates whether the true parameter is:

- A) Either larger or smaller than the hypothesized value
- B) Only smaller
- C) Only larger
- D) Equal

101. The significance level (alpha) is the probability of:

- A) Making a Type I error
- B) Making a Type II error
- C) The confidence level
- D) The effect size

102. Which test is used for a single proportion (large sample)?

- A) Chi-square test
- B) ANOVA
- C) z-test
- D) F-test

103. Comparing means from two related samples requires:

- A) Independent samples t-test
- B) Paired samples t-test
- C) Chi-square test
- D) ANOVA

104. An assumption of the classic independent-samples t-test is:

- A) Equal variances
- B) Unequal variances
- C) Zero variances
- D) Known variances

105. A Type II error means:

- A) Accepting H_0 when it is false
- B) Rejecting H_0 when it is true
- C) Accepting H_1 when it is true
- D) Rejecting H_1 when it is true

106. The F-test is used to:

- A) Compare two independent means
- B) Compare two variances
- C) Test independence
- D) Compare observed vs expected frequencies

107. Which test compares the variances of two normal populations?

- A) Chi-square test
- B) t-test
- C) F-test
- D) ANOVA

108. When testing a correlation coefficient, the usual H_0 is that:

- A) ρ is significantly positive
- B) ρ is significantly negative
- C) $\rho = 0$
- D) $\rho > 0$

109. The purpose of a flow diagram in hypothesis testing is to:

- A) Visually represent the steps in the process
- B) Calculate the test statistic
- C) Determine sample size
- D) Illustrate sample distribution

110. A limitation of hypothesis tests is that they:

- A) Provide effect size
- B) Apply only to large samples
- C) Do not measure practical significance
- D) Prove hypotheses definitively

111. Which test is typically used to compare more than two means?

- A) t-test
- B) ANOVA
- C) Chi-square test
- D) z-test

112. The chi-square test for variance is used to:

- A) Determine the population mean
- B) Compare sample means
- C) Compare a sample variance with a hypothesized population variance
- D) Test correlation

113. The chi-square test is considered non-parametric because it:

- A) Relies on population parameters
- B) Does not require distributional assumptions about the population
- C) Compares sample means
- D) Uses a normal distribution

114. Applying the chi-square test requires that the data be:

- A) Normally distributed
- B) Categorical
- C) Paired
- D) $n < 30$

115. For a chi-square goodness-of-fit test, H_0 states that:

- A) Observed and expected frequencies differ significantly
- B) Observed frequencies follow a specified distribution
- C) Variances are equal
- D) Sample means are different

116. Yates' continuity correction is used to:

- A) Increase the chi-square value
- B) Adjust the chi-square value in 2x2 tables with small expected counts
- C) Test for normality
- D) Adjust for multiple comparisons

117. The chi-square test for independence evaluates whether:

- A) Variances are equal
- B) Means are different
- C) Two categorical variables are related
- D) Two samples come from the same population

118. A primary reason to use sampling instead of a census is that sampling:

- A) Increases cost
- B) Reduces accuracy
- C) Is faster and less expensive
- D) Eliminates statistical analysis

119. Sampling is NOT appropriate when:

- A) The population is too large to study entirely
- B) A quick decision is required
- C) Every unit must be included
- D) Sampling costs less than a census

120. The term for the distribution of a sample statistic over all possible samples is:

- A) Sampling distribution
- B) Population distribution
- C) Sample space
- D) Distribution function

121. What is the main purpose of inferential statistics?

- A) To describe data
- B) To summarize data
- C) To infer population characteristics from a sample
- D) To collect data

122. Which of the following is true about the t-distribution?

- A) Used when sample size is large ($n > 30$)
- B) Used when population sigma is known
- C) Has heavier tails than the normal distribution
- D) Converges to a binomial distribution

123. The sampling distribution for proportions in large samples is:

- A) Normal distribution
- B) Chi-square distribution
- C) Poisson distribution
- D) Exponential distribution

124. According to the Central Limit Theorem, the sampling distribution of the sample mean:

- A) Approaches a uniform distribution
- B) Approaches a binomial distribution
- C) Approaches a normal distribution as n increases
- D) Becomes identical to the population distribution

125. A key implication of the Central Limit Theorem is that:

- A) The mean of sample means exceeds the population mean
- B) The variance of sample means exceeds the population variance
- C) The distribution of sample means approaches normality even if the population is not normal
- D) The sample mean always equals the population mean

126. Simple random sampling means:

- A) Every population member has an equal chance of selection
- B) Samples are chosen by specific characteristics
- C) Samples are taken at fixed intervals
- D) Samples are pre-grouped then selected

127. Stratified sampling means:

- A) Population divided into clusters and one cluster sampled
- B) Every member has an equal chance of selection
- C) Population divided into strata and random samples taken from each
- D) Samples taken at regular intervals

128. Sandler's A-test assesses:

- A) Variance within one sample
- B) Difference between two variances
- C) Agreement between two means
- D) Magnitude of measurement errors in repeated measures

129. An assumption of Sandler's A-test is that:

- A) Data are normally distributed
- B) Sample sizes are equal
- C) Samples are dependent
- D) Population variance is known

130. The standard error of the mean measures:

- A) Dispersion of data around the sample mean
- B) Variability of the sample mean relative to the population mean
- C) Variability of individual observations
- D) Central tendency of the population

131. Which factor affects the standard error of the mean?

- A) Population mean
- B) Sample size
- C) Population median
- D) Sample median

132. A point estimate is:

- A) An interval estimate providing a range
- B) A single-value estimate of a population parameter
- C) Always equal to the population parameter
- D) A biased estimate

133. A property of a good estimator is:

- A) Bias
- B) Efficiency
- C) Variability
- D) Complexity

134. When estimating a population mean with small n and unknown sigma, use:

- A) Normal distribution
- B) t-distribution
- C) Binomial distribution
- D) Poisson distribution

135. In the confidence-interval formula $\bar{x} \pm t * s/\sqrt{n}$, the t represents:

- A) Critical value from the normal distribution
- B) Critical value from the t-distribution
- C) Sample mean
- D) Standard deviation

136. The sample proportion $\hat{p}$ is:

- A) Always equal to the population proportion
- B) A point estimate of the population proportion**
- C) Always biased
- D) Constant across samples

137. In the margin-of-error formula $\pm z\sqrt{p(1-p)/n}$, the z represents:

- A) Sample proportion
- B) Standard deviation
- C) Critical value from the normal distribution**
- D) Mean of sample proportions

138. Which factor does not directly affect required sample size?

- A) Desired margin of error
- B) Confidence level
- C) Population size**
- D) Population variance

139. When computing sample size using precision rate and confidence level, which is not required?

- A) Desired confidence level
- B) Population size**
- C) Precision rate (margin of error)
- D) Population variance

140. To obtain a narrower confidence interval at the same confidence level, sample size should be:

- A) Increased**
- B) Decreased
- C) Kept constant
- D) Doubled

141. Bayesian sample-size determination incorporates:

- A) Prior and posterior distributions**
- B) Confidence intervals only
- C) Sample mean alone
- D) Sample proportion alone

142. A characteristic of Bayesian sample-size determination is:

- A) It ignores prior information
- B) Uses maximum-likelihood estimates only
- C) Incorporates prior beliefs and evidence**
- D) Independent of the prior distribution

143. For a skewed population, the CLT generally holds if $n \geq$:

- A) 10
- B) 20

C) 30

D) 50

144. The main advantage of stratified sampling over simple random sampling is:

A) Easier to administer

B) Reduces sampling error by representing all subgroups

C) Requires fewer resources

D) Simpler analysis

145. The first step in the research process is:

A) Reviewing literature

B) Defining the research problem

C) Collecting data

D) Formulating a hypothesis

146. A research problem should be:

A) Complex and unstructured

B) Clear and precise

C) Vague and general

D) Theoretical only

147. In medical research, informed consent is essential because it:

A) Ensures researcher control

B) Protects legal interests

C) Respects participants' autonomy and rights

D) Simplifies procedures

148. Which ethical principle means "do no harm"?

A) Beneficence

B) Non-maleficence

C) Justice

D) Autonomy

149. Which research design establishes cause-and-effect relationships?

A) Descriptive

B) Experimental

C) Correlational

D) Exploratory

150. A case study is an example of:

A) Quantitative research

B) Qualitative research

C) Experimental research

D) Survey research

151. The primary goal of sampling is:

A) To reduce cost

- B) To survey all population members
- C) To generalize findings from a sample to a population
- D) To simplify data entry

152. Which sampling method ensures equal selection probability?

- A) Stratified sampling
- B) Cluster sampling
- C) Systematic sampling
- D) Simple random sampling

153. Which measurement scale classifies data into distinct categories with no order?

- A) Ordinal
- B) Nominal
- C) Interval
- D) Ratio

154. An example of an interval scale is:

- A) Temperature in degrees C
- B) Height in cm
- C) Number of children
- D) Gender

155. Which data-collection method is typical for qualitative research?

- A) Surveys
- B) Interviews
- C) Questionnaires
- D) Checklists

156. Secondary data are:

- A) Collected firsthand by the researcher
- B) Obtained from existing sources
- C) Collected through experiments
- D) Gathered via surveys

157. Transforming raw data into meaningful information is called:

- A) Data cleaning
- B) Data processing
- C) Data collection
- D) Data storage

158. A common data-processing step is:

- A) Data collection
- B) Data entry
- C) Data analysis
- D) Dissemination

159. A sampling frame is:

- A) A storage device for samples
- B) A list of all items in the population
- C) A selection method
- D) A non-random technique

160. Dividing the population into subgroups and sampling each defines:

- A) Simple random sampling
- B) Cluster sampling
- C) Stratified sampling
- D) Systematic sampling

161. Which test compares means of two independent groups?

- A) Chi-square test
- B) t-test
- C) ANOVA
- D) Mann-Whitney U test

162. A p-value < 0.05 indicates:

- A) Strong evidence for H0
- B) Strong evidence against H0
- C) No evidence against H0
- D) Data error

163. The chi-square test is used to:

- A) Compare means
- B) Test independence between categorical variables
- C) Compare variances
- D) Analyze variance

164. The chi-square test applies when data are:

- A) Ordinal
- B) Nominal
- C) Interval
- D) Ratio

165. ANOVA is used to:

- A) Compare two means
- B) Compare three or more means
- C) Compare two variances
- D) Compare medians

166. Which technique adjusts means for covariates?

- A) ANOVA
- B) ANCOVA
- C) MANOVA
- D) MANCOVA

167. Which non-parametric test compares two related samples?

- A) Wilcoxon signed-rank test
- B) Mann-Whitney U test
- C) Kruskal-Wallis test
- D) Friedman test

168. The Mann-Whitney U test compares:

- A) Means of two independent samples
- B) Medians of two independent samples
- C) Means of two related samples
- D) Medians of two related samples

169. Which technique identifies underlying factors among correlated variables?

- A) Factor analysis
- B) Cluster analysis
- C) Discriminant analysis
- D) Regression analysis

170. Principal Component Analysis (PCA) is primarily used to:

- A) Classify data
- B) Reduce data dimensionality
- C) Predict outcomes
- D) Test hypotheses

171. Which section of a research report presents main findings?

- A) Introduction
- B) Literature review
- C) Results
- D) Discussion

172. A well-written research report should be:

- A) Lengthy and complex
- B) Concise and clear
- C) Filled with jargon
- D) Only positive

173. The primary role of computers in research is:

- A) To replace researchers
- B) To assist in data collection and analysis
- C) To complicate research
- D) To eliminate all errors

174. Which software is widely used for statistical analysis?

- A) Microsoft Word
- B) Adobe Photoshop
- C) SPSS
- D) AutoCAD

175. Which Stata command summarizes data?

- A) summarize
- B) tabulate
- C) regress
- D) correlate

176. In Excel, which function calculates the mean of a range?

- A) MEDIAN
- B) MODE
- C) AVERAGE
- D) SUM

177. The main objective of medical research is:

- A) To confirm personal opinions
- B) To generate new knowledge and improve health outcomes
- C) To repeat old experiments
- D) To promote treatments

178. Research design is best defined as:

- A) Random data collection
- B) A detailed plan for conducting the study
- C) A hypothesis statement
- D) Literature review

179. In research, a variable is:

- A) A constant value
- B) A characteristic that can vary
- C) An unmeasurable factor
- D) A controlled setting

180. Which is NOT a step of the scientific method?

- A) Defining the problem
- B) Data collection
- C) Guessing without analysis
- D) Drawing conclusions

181. The purpose of a conceptual framework is:

- A) To visualize relationships between variables
- B) To replace statistics
- C) To summarize results
- D) To prepare ethics approval

182. Which type of research applies results directly in practice?

- A) Basic research
- B) Applied research
- C) Theoretical research
- D) Exploratory research

183. Which of the following is a quantitative variable?

- A) Blood-pressure value
- B) Blood-group type
- C) Marital status
- D) Country of origin

184. Which measurement scale has a true zero point?

- A) Nominal
- B) Ordinal
- C) Interval
- D) Ratio

185. The primary purpose of pilot studies is:

- A) To finalize conclusions
- B) To test tools and feasibility before the main study
- C) To increase sample size
- D) To replace data collection

186. Which sampling technique uses a fixed interval between elements?

- A) Simple random sampling
- B) Systematic sampling
- C) Cluster sampling
- D) Snowball sampling

187. An extraneous variable is:

- A) Controlled by the researcher
- B) Influences the dependent variable but is not of interest
- C) The main study variable
- D) A constant factor

188. Data coding means:

- A) Confusing results
- B) Converting raw data into numerical/symbolic form
- C) Hiding errors
- D) Summarizing theories

189. Data accuracy during entry is ensured by:

- A) Cross-checking and validation
- B) Ignoring missing values
- C) Random typing
- D) Deleting incomplete surveys

190. A study that follows participants forward in time is:

- A) Retrospective study
- B) Cross-sectional study
- C) Cohort study
- D) Case-control study

191. In a case-control study, participants are selected by:

- A) Exposure status
- B) Outcome or disease status**
- C) Random assignment
- D) Age category

192. Which test compares frequencies between two categorical variables?

- A) t-test
- B) ANOVA
- C) Chi-square test**
- D) Correlation analysis

193. The p-value represents:

- A) Sample mean
- B) Probability of obtaining results by chance**
- C) Number of variables
- D) Level of reliability

194. A confidence interval indicates:

- A) Range where the population parameter likely lies**
- B) Fixed standard deviation
- C) Variable's significance
- D) Title relevance

195. Sampling bias is reduced by:

- A) Convenience sampling
- B) Random sampling**
- C) Quota sampling
- D) Purposive sampling

196. A research proposal is:

- A) Final paper
- B) Preliminary plan outlining purpose, methods, and expected outcomes**
- C) Survey result
- D) Statistical test

197. Reliability of a questionnaire can be measured using:

- A) Cronbach's alpha**
- B) Regression coefficient
- C) Mean-square error
- D) Skewness value

198. The ethical principle of beneficence means:

- A) To gain financial benefit
- B) To maximize benefits and minimize harm**
- C) To share data publicly
- D) To speed research

199. Which document defines ethical standards for human research?

- A) Nuremberg Code
- B) Civil Code
- C) Criminal Code
- D) Education Law

200. The purpose of randomization in experiments is:

- A) Assign by preference
- B) Ensure comparable groups and reduce bias
- C) Shorten the study
- D) Increase sample size

201. Which type of error occurs when a false null hypothesis is accepted?

- A) Type I error
- B) Type II error
- C) Sampling error
- D) Measurement error

202. What is the role of peer reviewers in publication?

- A) To write articles for authors
- B) To evaluate the quality and validity of manuscripts
- C) To promote journals
- D) To manage finances

203. The abbreviation SPSS stands for:

- A) Statistical Package for the Social Sciences
- B) Systematic Program for Statistical Studies
- C) Simple Processing Software System
- D) Sample Population Statistical Sheet

204. In Stata, which command is used to import Excel data?

- A) read.csv
- B) import excel
- C) loadfile
- D) use excel

205. Which graphical tool shows the distribution of one quantitative variable?

- A) Pie chart
- B) Histogram
- C) Scatter plot
- D) Bar chart

206. What is the main function of a research abstract?

- A) To present detailed data
- B) To summarize the entire study briefly
- C) To list references
- D) To describe only methodology

207. Which referencing style is used most often in biomedical journals?

- A) APA
- B) Vancouver**
- C) Chicago
- D) MLA

208. What is the main difference between qualitative and quantitative research?

- A) Qualitative focuses on numbers; quantitative focuses on meanings
- B) Qualitative focuses on meanings; quantitative focuses on numbers**
- C) Both focus on statistical inference
- D) There is no difference

209. Which research method emphasizes understanding participants' experiences?

- A) Experimental study
- B) Qualitative interview**
- C) Cross-sectional survey
- D) Regression analysis

210. In research, what does validity mean?

- A) Consistency of measurement
- B) Accuracy of measurement**
- C) Randomness of measurement
- D) Complexity of instrument

211. What is a dependent variable?

- A) The variable that is manipulated
- B) The variable that is affected by changes in another variable**
- C) A constant factor
- D) A random element

212. What is the purpose of inclusion criteria in research?

- A) To exclude participants
- B) To define which participants qualify to enter the study**
- C) To randomize data
- D) To collect secondary data

213. The process of selecting participants who meet certain characteristics is called:

- A) Randomization
- B) Sampling
- C) Recruitment**
- D) Stratification

214. Which sampling technique selects groups rather than individuals?

- A) Simple random sampling
- B) Cluster sampling**
- C) Stratified sampling
- D) Systematic sampling

215. What is a research hypothesis?

- A) A proven fact
- B) A tentative explanation or prediction that can be tested
- C) A random assumption
- D) A statistical method

216. What is the main purpose of data analysis?

- A) To collect data
- B) To organize, interpret, and draw conclusions from data
- C) To delete outliers
- D) To visualize results only

217. Which type of study examines data from different groups at one point in time?

- A) Cohort study
- B) Cross-sectional study
- C) Experimental study
- D) Longitudinal study

218. What is a research gap?

- A) An unanswered or underexplored question in existing studies
- B) A statistical error
- C) A missing data point
- D) A mistake in sampling

219. The process of checking the accuracy and consistency of collected data is called:

- A) Validation
- B) Classification
- C) Tabulation
- D) Quantification

220. In research, triangulation is used to:

- A) Test normality of data
- B) Use multiple sources or methods to confirm findings
- C) Measure reliability
- D) Simplify analysis

221. What is the main purpose of literature citation?

- A) To show reading habits
- B) To give credit to original authors and avoid plagiarism
- C) To fill space
- D) To support personal opinions

222. What is bias in research?

- A) Random variation in data
- B) Systematic error that affects validity
- C) Measurement error due to instruments
- D) Missing data in the dataset

223. What does placebo mean in a clinical trial?

- A) An inactive substance used to compare with a treatment
- B) A strong experimental drug
- C) A harmful chemical
- D) An independent variable

224. In medical ethics, autonomy refers to:

- A) The researcher's independence
- B) The participant's right to make informed decisions
- C) The doctor's authority
- D) Institutional control

225. What is the purpose of the Declaration of Helsinki?

- A) To define national research funding
- B) To regulate ethical principles for medical research involving humans
- C) To measure validity
- D) To approve budgets

226. What does confidentiality mean in research ethics?

- A) Sharing data with all colleagues
- B) Protecting participants' personal information
- C) Publishing all names
- D) Allowing sponsors to see raw data

227. Which document is typically submitted to obtain ethical approval?

- A) Research proposal
- B) Informed consent form
- C) Ethics application with full protocol
- D) Literature review

228. Difference between primary and secondary data:

- A) Primary is collected directly; secondary from existing sources
- B) Primary is old; secondary is new
- C) Primary is from books; secondary from surveys
- D) Primary is theoretical; secondary practical

229. Which statistical test analyzes the relationship between two continuous variables?

- A) Chi-square test
- B) Correlation test
- C) ANOVA
- D) Mann-Whitney U test

230. Standard deviation is:

- A) The square root of variance
- B) The average of all data
- C) The total number of cases
- D) Difference between means

231. A research instrument is reliable if it:

- A) Produces consistent results under the same conditions
- B) Always gives higher values
- C) Measures multiple variables simultaneously
- D) Involves random errors

232. Data visualization is used to:

- A) Improve text structure
- B) Present data clearly via graphs and charts
- C) Delete missing values
- D) Perform correlation

233. Which graph is best for showing proportions?

- A) Bar chart
- B) Pie chart
- C) Histogram
- D) Line chart

234. The role of sample-size calculation is:

- A) To guarantee adequate statistical power for detecting meaningful effects
- B) To minimize the time required to perform analytical procedures in research
- C) To intentionally introduce bias during participant or data selection processes
- D) To make scientific manuscript writing simpler without methodological rigor

235. The function of the SPSS Output Viewer is:

- A) To generate complete narrative reports without human interpretation
- B) To display outcomes of statistical analyses for review and interpretation
- C) To digitally collect and store completed survey questionnaires from respondents
- D) To perform structured coding of raw dataset values prior to input processing

236. In Stata, the command summarize varname provides:

- A) A visual graph illustrating distribution patterns across variables
- B) Descriptive statistical indicators related to the selected variable
- C) A generated output containing multi-level regression model results
- D) A comprehensive matrix examining correlation among multiple variables

237. An advantage of using Stata over Excel in data analysis is:

- A) The ability to format tables and visuals more easily for presentation purposes
- B) The availability of advanced statistical tools with high analysis reproducibility
- C) Integration of automated literature review from scientific databases
- D) Inclusion of a built-in module for composing detailed research manuscripts

238. What is a manuscript cover letter?

- A) A standardized template used exclusively for manual data entry into software
- B) A formal letter submitted to the editor to introduce and justify the article
- C) An evaluative feedback document provided by a peer reviewer during assessment
- D) An informed consent form signed by participants taking part in the investigation

239. RCT stands for:

- A) Random Case Trial
- B) Randomized Controlled Trial
- C) Retrospective Clinical Test
- D) Repeated Control Testing

240. Main feature of a double-blind study:

- A) Both researcher and participant know treatment assignment
- B) Only participants know
- C) Neither researcher nor participant knows
- D) Both groups use placebos

241. Which of the following is an example of descriptive statistics?

- A) Mean, median, mode
- B) Regression analysis
- C) ANOVA
- D) Chi-square test

242. The interquartile range (IQR) measures:

- A) The spread of the middle 50% of data
- B) The mean difference
- C) The total data range
- D) The mode of a dataset

243. Main goal of data cleaning is:

- A) To remove errors and inconsistencies from data
- B) To increase dataset size
- C) To generate hypotheses
- D) To simplify graphs

244. The null hypothesis (H₀) states that:

- A) There is no significant relationship or difference
- B) There is a strong relationship
- C) The results are biased
- D) The experiment is invalid

245. The alternative hypothesis (H₁) suggests that:

- A) There is no effect
- B) There is an effect or relationship
- C) Variables are identical
- D) Data are invalid

246. A research population is:

- A) The entire group relevant to the study
- B) A tested subset
- C) A list of data points
- D) A statistical summary

247. Random sampling is used to:

- A) Minimize selection bias
- B) Increase the number of variables
- C) Test only volunteers
- D) Simplify analysis

248. A cross-over design is:

- A) A study where each participant receives several interventions sequentially
- B) A research method based solely on observational data collection techniques
- C) An experimental approach applying only a double-blind placebo methodology
- D) A statistical process used to combine results from multiple published studies

249. Data tabulation means:

- A) The systematic process of organizing information into structured tables
- B) The removal of all missing values from the original research database
- C) Application of statistical hypothesis testing to confirm assumed results
- D) Classification of randomly observed variables with minimal organization

250. Informed consent includes:

- A) Providing research details, risks, benefits, and participant legal rights
- B) Sharing financial compensation and payment information with respondents
- C) Presenting formal academic qualifications of the principal investigator
- D) Stating deadlines and official time limits related to study implementation

251. A main advantage of standardized questionnaires is:

- A) Ensuring comparability of collected results across different research studies
- B) Increasing the likelihood of excluding ethical oversight requirements
- C) Eliminating the necessity of obtaining participant informed consent
- D) Enhancing the amount of unintended bias introduced during data collection

252. In ANOVA, the F-statistic is the ratio of:

- A) Two unrelated sample variances that are computed independently
- B) Variance between groups divided by variance within groups
- C) Arithmetic mean values calculated from two distinct populations
- D) A covariance measurement illustrating variable interdependence

253. Data normalization is:

- A) Scaling values to a common range without affecting meaningful differences
- B) Eliminating every detected outlier from the dataset without further review
- C) Rounding each data point to a reduced number of decimal places
- D) Transforming quantitative data values into categorical (qualitative) form

254. The median is:

- A) The observation that occurs most frequently within a given dataset
- B) The central value when all data are arranged in ascending or descending order
- C) The smallest numerical value identified in the collection of observations
- D) The total sum of values divided by the count of data points collected

255. Standard error (SE) indicates:

- A) The variability of the sample mean compared to the true population mean
- B) The numerical difference between calculated median and mode values
- C) The level of error attributed to imperfections in the measurement tool
- D) The directional bias present during selection of research sample participants

256. Which test is non-parametric?

- A) ANOVA
- B) Mann-Whitney U test
- C) t-test
- D) z-test

257. Multicollinearity in regression means:

- A) Independent variables are highly correlated
- B) Dependent variable is missing
- C) Residuals are random
- D) Outliers are removed

258. Heteroscedasticity is:

- A) Unequal variance of residuals
- B) Equal variance across groups
- C) Perfect correlation
- D) Missing-variable bias

259. To assess the strength of a linear relationship, use:

- A) Mean
- B) Correlation coefficient (r)
- C) Standard deviation
- D) Median

260. $p < 0.05$ usually indicates:

- A) Results due to chance
- B) Statistically significant results
- C) Invalid results
- D) Sample too small

261. The best chart for showing time trends is a:

- A) Pie chart
- B) Bar chart
- C) Line chart
- D) Histogram

262. Best research method for rare diseases:

- A) Cross-sectional study
- B) Case-control study
- C) Cohort study
- D) Experimental design

263. External validity means:

- A) Generalizability to other populations or settings
- B) Accuracy of measurement tools
- C) Random error control
- D) Internal consistency

264. A data-coding sheet contains:

- A) Variables, labels, and assigned numerical codes
- B) Names of researchers
- C) Research timeline
- D) Interview transcripts

265. A scatter plot is used for:

- A) Displaying the relationship between two quantitative variables
- B) Comparing the frequencies of multiple study groups
- C) Showing proportional distributions of categorical data
- D) Demonstrating the ranked order of observed outcomes

266. A control variable is:

- A) A variable intentionally kept constant to avoid influencing the results
- B) A variable that is actively manipulated by the researcher during the study
- C) A variable that changes randomly without systematic control
- D) A variable that depends entirely on other measured variables

267. An example of secondary data is:

- A) Data collected from medical records
- B) Data collected by a survey
- C) Lab test results
- D) Patient observations

268. The purpose of research ethics committees is:

- A) To approve financial resources for research projects
- B) To verify that studies comply with ethical and safety regulations
- C) To conduct statistical examination of collected study results
- D) To carry out routine administrative data processing activities

269. In epidemiology, incidence refers to:

- A) The complete number of already existing disease cases in a population
- B) The count of newly occurring disease cases within a defined time
- C) The total number of individuals who are considered at health risk
- D) The measurement of symptom frequency across different subjects

270. Prevalence means:

- A) The total number of new disease cases occurring during a study period
- B) The complete number of existing cases at a specific time point
- C) The total mortality rate observed in the studied population
- D) The total number of clinically recovered patients reported

271. Meta-analysis is:

- A) A structured review statistically combining results from several studies
- B) A research project based on a single observational investigation
- C) A qualitative data collection technique using guided interviewing
- D) A descriptive summary report without statistical interpretation

272. Publication bias is:

- A) The tendency to publish mainly studies with statistically significant outcomes
- B) A random computational mistake occurring during data analysis procedures
- C) The use of incorrect citation practices in preparing the manuscript
- D) A time delay caused by reviewers during the evaluation process

273. Gray literature refers to:

- A) Non-commercial or unpublished research such as theses or institutional reports
- B) Peer-reviewed scientific articles featured in highly ranked journals
- C) Promotional marketing material intended for public advertisement
- D) Research posters presented solely at scientific conference events

274. A prospective study:

- A) Monitors selected individuals moving forward in time to observe outcomes
- B) Investigates previously recorded information from hospital documentation
- C) Reviews and assesses already published scientific literature
- D) Utilizes group discussion sessions to collect qualitative perspectives

275. A retrospective study:

- A) Collects data going forward in time
- B) Analyzes existing or past data
- C) Conducts lab experiments
- D) Predicts future results

276. The main goal of a systematic review is:

- A) To summarize and synthesize results from multiple studies objectively
- B) To conduct interviews
- C) To create new experiments
- D) To publish personal opinions

277. An operational definition is:

- A) A precise and measurable description of how a variable will be assessed
- B) A general written summary of literature sourced from scientific databases
- C) A broad theoretical framework explaining abstract study concepts
- D) A simple graphical diagram representing study components visually

278. Pilot testing of instruments ensures:

- A) Practical feasibility, measurement reliability, and clarity of data tools
- B) Automatic participant allocation through a randomization procedure
- C) Immediate ethical approval from relevant institutional review authority
- D) Achievement of statistical significance for all measured study outcomes

279. Data triangulation is:

- A) The use of several independent data sources to confirm research findings
- B) Construction of exactly three analytical variables within the data
- C) Repeated utilization of a single questionnaire tool without modifications
- D) Manual application of random allocation techniques during experimentation

280. Construct validity is:

- A) The extent to which an instrument accurately measures the intended concept
- B) The practical capability to reproduce identical research results repeatedly
- C) The degree of variability observed across multiple study outcome values
- D) The accuracy level of input during manual data entry processes

281. Which statistical test examines relationships between categorical variables?

- A) Pearson correlation
- B) Chi-square test
- C) t-test
- D) Regression analysis

282. Which test compares three or more means?

- A) ANOVA
- B) t-test
- C) Chi-square
- D) Correlation

283. Regression analysis aims to determine:

- A) The relationship between dependent and independent variables
- B) The average of data points
- C) The validity of questionnaires
- D) The proportion of missing data

284. Graphically, the normal distribution appears as a:

- A) Bell-shaped curve
- B) Skewed curve
- C) Flat line
- D) Zigzag pattern

285. Skewness indicates:

- A) The asymmetry of a data distribution
- B) The variability of variance
- C) The number of observations
- D) The mean value

286. Flattening or peaking of a distribution refers to:

- A) Kurtosis
- B) Variance
- C) Dispersion
- D) Normality

287. Confidence level means:

- A) The probability that the interval contains the true population parameter
- B) The total number of samples included in the current investigation
- C) The correlation coefficient calculated from the study variables
- D) The numeric value of the statistical test statistic used in analysis

288. A common confidence level in research is:

- A) Fifty percent of the measurement accuracy reported in analysis
- B) Ninety percent of the commonly accepted statistical certainty level
- C) Ninety-five percent reliability used in most scientific publications
- D) Seventy-five percent reliability applied for preliminary assessments

289. Effect size is:

- A) The magnitude of difference or association observed between variables
- B) The arithmetic mean derived from the collected sample data
- C) The calculated probability of committing statistical error in testing
- D) The measured variance computed for the available research sample

290. A data entry error is:

- A) Incorrect recording of data values during the input phase
- B) Absence of selected respondents from the participation process
- C) Partial execution of required literature review activities
- D) Violation of ethical standards during research implementation

291. In clinical trials, an adverse event refers to:

- A) Any undesired medical occurrence arising in a study participant
- B) A beneficial and expected outcome following an intervention
- C) Random computational mistake within statistical analysis software
- D) Official rejection of the trial during the ethical review procedure

292. The function of a Data Monitoring Committee (DMC) is:

- A) To supervise safety and scientific validity throughout clinical trials
- B) To manage participant recruitment and enrollment processes
- C) To prepare scientific manuscripts intended for journal submission
- D) To collect and label biological samples for laboratory evaluation

293. Dropout rate means:

- A) Percentage of participants who discontinue involvement before study end
- B) Total number of individuals initially enrolled in the investigation
- C) Statistical deviation introduced during the process of sample selection
- D) Quantity of research variables excluded from the analytical procedure

294. Attrition bias is:

- A) Systematic distortion caused by unequal participant loss between groups
- B) Error resulting from faulty random allocation of study participants
- C) Bias produced due to incorrect or inconsistent measurement tools
- D) Selective publication of studies showing statistically significant results

295. A data confidentiality breach is:

- A) Unauthorized disclosure or access to participants' private information
- B) Failure to obtain required data from certain study respondents
- C) Incorrect application of coding principles during data processing
- D) External contamination of biological samples used in analysis

296. Repeating a study to confirm results is called:

- A) Verification
- B) Replication
- C) Validation
- D) Randomization

297. The introduction of a research paper should:

- A) Present study background, problem statement, and research objectives
- B) Provide references with links retrieved directly from Scopus database records
- C) Display graphical illustrations without additional explanatory text
- D) Offer a brief overview of descriptive statistical outcomes only

298. The methods section of a manuscript should:

- A) Explain in detail how the research was structured and executed
- B) Present analytical outcomes obtained from data processing
- C) Provide a narrative summary of general study interpretations and results
- D) Include a complete glossary of abbreviations used throughout the manuscript

299. The results section should primarily contain:

- A) Objective factual findings presented clearly without author interpretation
- B) Explanatory evaluation of study outcomes compared with earlier research work
- C) Citations and references supporting the theoretical background discussion
- D) Expressive personal viewpoints of the researcher on study performance

300. The discussion section should:

- A) Interpret results, compare with previous studies, and highlight implications
- B) Present only unformatted raw numerical data collected during the study
- C) Repeat detailed methodology described earlier in the manuscript sections
- D) Contain solely tables without explanatory narrative or contextual relevance